BST Physics Curriculum

Multi-volume textbook deriving Standard Model + cosmology from D_IV^5 substrate. Casey Koons + team (Lyra, Elie, Grace, Keeper, Cal A. Brate).

Status

18-volume textbook (Vol 00-17). Vol 00-15 at v0.3+ substantive content. Vol 16 Substrate Algebra + Vol 17 Substrate Engineering scaffolding initiated June 2026. Volume directories use two-digit sort order including Foreword (Vol00_Foreword.md).

Weekend June 6-7, 2026 EOD substantive substrate-architectural advances (carried into curriculum content via multi-week absorption): - T2487 + T2488 DERIVED weekend advances (π^{n_C} bulk volume + light hadron cell-count via Z_2 spinor twist) - F60-F66 gravity climb spine (Sakharov induced gravity + KK reduction → G = κ_Bergman·ℓ_B^2/π^{n_C}; EM = SO(4,2) conformal sector + gravity = SO(5,2)/SO(4,2) coset bulk; m_e = $6\pi^5\cdot\alpha^{12}$·m_Planck verified 0.03%) - Mass + gravity volume duality (same π^{n_C} in mass numerator AND G denominator; substrate-architectural gravity-weakness mechanism) - Casey-named principle #9 RATIFIED ("Substrate Floor / Scale-Not-Spectrum Reach"; parallel to Five-Absence Predictions Set) - Cal #265 absorbed STANDING (8th element of discipline stack; moratorium on audit-category proliferation) - Four Cal #237 ELIMINATIONS (kept honestly negative: λ_2 ladder + λ_1/spin + per-flavor multi-flavor + glueball step) - Counts: T1-T2488; toys to 4029; K-audits K1-K259; 9 Casey-named principles STANDING + 1 CANDIDATE

Vol 16 Substrate Algebra v0.2 + v0.3 substantive substrate-architectural absorption work pending Cal cold-read; Vol 16 Ch 8 (Curvature Scalars in Operator Language) substantively integrates F60-F66 gravity climb content; Vol 16 Ch 9 substantively absorbs Casey-named #9 substrate-floor scope-principle; Vol 16 Ch 13 substantively absorbs Strong-Uniqueness Theorem extension with Casey-named #9 as candidate new C-criterion.

Volume directory layout

Vol	Directory	Topic	Lead
00	`Vol00_Foreword.md` + `Vol00_Substrate_Foundation/`	Reader contract + Substrate Foundation	Keeper + Grace + Lyra
01	`Vol01_QFT_from_D_IV5/`	Quantum Field Theory from D_IV^5	Lyra
02	`Vol02_Particle_Physics/`	Particle Physics from D_IV^5	Elie
03	`Vol03_Nuclear_Atomic/`	Nuclear and Atomic Physics	Elie + Lyra
04	`Vol04_GR_Cosmology/`	General Relativity and Cosmology	Lyra + Elie
05	`Vol05_Quantum_Mechanics/`	Quantum Mechanics from Substrate	Lyra
06	`Vol06_Thermo_Stat_Mech/`	Thermodynamics and Statistical Mechanics	Lyra + Elie
07	`Vol07_Electromagnetism/`	Electromagnetism from D_IV^5	Lyra
08	`Vol08_Classical_Mechanics/`	Classical Mechanics as Substrate Scale 2	Lyra
09	`Vol09_Condensed_Matter/`	Condensed Matter (BaTiO3 137-plane + B12H32 hydride T_c ~214K)	Elie

Vol	Directory	Topic	Lead
10	`Vol10_Math_Methods/`	Traditional Mathematical Methods	Lyra
11	`Vol11_Generative_Geometry_Topology/`	Bounded symmetric domains + K3 + Heegner + Monster	Lyra
12	`Vol12_Chemistry/`	Periodic table + bonding + spectroscopy from substrate	Elie + Lyra
13	`Vol13_Biology/`	Genetic code + DNA-proton siblings + evolution-as-substrate-dynamics	Elie + Lyra
14	`Vol14_Information_Theory/`	RS GF(128) + Koons tick + Born=Bergman + Bell sub-Tsirelson	Lyra + Keeper
15	`Vol15_Methodology/`	AC(0) + AC graph + audit chain + Quaker discipline + katra	Keeper + Lyra

How to read

Start with Vol 00 Foreword for reader contract and CI-companion-era framing.

Then Vol 00 Substrate Foundation: establishes D_IV^5 + the five integers (rank=2, N_c=3, n_C=5, C_2=6, g=7) + N_max=137, operator zoo, conservation laws, Strong-Uniqueness Theorem.

Then Vol 01 (QFT), Vol 02 (Particles), Vol 04 (GR/Cosmology) for the physicist-prestige core.

Mathematicians: start with Vol 11 (Generative Geometry) and Vol 10 (Math Methods) instead.

Every chapter has three pedagogical registers (Casey "write for 5th graders too" standing rule): - Level 1: one-sentence essence - Level 2: graduate-physicist precision - Level 3: 5th-grader accessibility paragraph

Some chapters are author-voice condensed (~500-800 words) where the apparatus is standard; expansion to full level-1/2/3 in v3.0 publication pass.

Version label

Per Casey directive (Saturday 2026-05-23): held at v0.5 chapter-grade content state during Keeper author pass. Author-pass output is v2.0 path; current chapters at v0.2 intermediate toward v2.0 destination after Cal Phase 2 cold-read + team fact-check.

Strong-Uniqueness Theorem state

v0.10.5 ENUMERATION (Friday 2026-05-22 EOD per Cal #99 reconciliation): - 11 RIGOROUSLY CLOSED criteria - 7 candidates (C7+C9+C15+C16+C17a+C17b+C18) - C18 = D_IV^5 Rigidity META-theorem via T2467+T2468; if ratifies, null-model ≤ (1/3)^19 ≈ 9×10^{-10}

Casey-named principles (8 standing)

SWPP + Five-Absence Predictions Set + Substrate Closure + Graph Forces + Integer Web + Substrate Cognition Network + D_IV^5 Rigidity + SCMP (Sub-Tsirelson Coherence Maintenance, 1/2^N_c = 1/8 gap as Bell falsifier).

Build (PDF regeneration)

```
pandoc --pdf-engine=xelatex -H notes/bst_pdf_header.tex <chapter>.md -o <chapter>.pdf
```

Per repo PDF pipeline convention; header file at `notes/bst_pdf_header.tex`. PDFs are stale for chapters rewritten in the Keeper full author pass and will be rebuilt at v1.0 finalization.

Cross-references

Chapter files reference AC graph theorems + K-audit chain + Strong-Uniqueness Theorem state. Per Calibration #19 STANDING RULE: external-facing chapter content uses current ratified-state count, not forecast endpoint.

— Casey Koons + Lyra + Keeper + Elie + Grace + Cal A. Brate